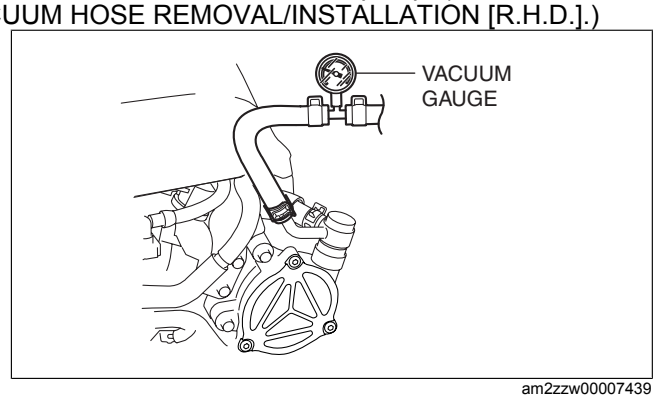
VACUUM PUMP INSPECTION

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SKYACTIV-G 1.3, SKYACTIV-G 1.5

1. Pinch open the clamp using pliers and disconnect the vacuum hose from the vacuum pump. (See VACUUM HOSE REMOVAL/INSTALLATION [L.H.D.].) (See VACUUM HOSE REMOVAL/INSTALLATION [R.H.D.].)
2. Install the vacuum gauge.
3. Warm up the engine.
4. Switch the ignition off.
5. Depress the brake pedal several times and decrease the vacuum of the vacuum gauge to **0 kPa {0 mmHg, 0 inHg}**.
6. Switch the ignition ON (engine on).



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Caution

- **If the brake pedal is released after switching the ignition ON (engine on), the vacuum in the vacuum pump may decrease and result in an incorrect reading when performing the inspection for the degree of vacuum in the vacuum pump. Depress and hold the brake pedal after switching the ignition ON (engine on).**

7. Perform the inspection for the degree of vacuum in the vacuum pump.
   (1) Measure the time from the measurement-start vacuum to the measurement-stop vacuum while taking readings on the vacuum gauge.

Degree of vacuum in vacuum pump standard (reference) [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

Measurement condition				Specification
Engine rotation speed	Brake pedal	Measurement-start vacuum	Measurement-stop vacuum	
While idling (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)	Depressed	60 kPa {450 mmHg, 18 inHg}	67 kPa {503 mmHg, 20 inHg}	Within 8 s

   (2) Verify that the measured time matches the standard.
   - If they do not match, repair or replace the following malfunctioning locations.
     — Vacuum pump malfunction
     — Engine oil pressure decrease
   - If they match, go to the next step.

8. Perform the inspection for the degree of maximum attained vacuum in the vacuum pump.

Note

- The inspection for the degree of maximum attained vacuum in the vacuum pump can be performed normally even if the brake pedal is released.

   (1) Read the vacuum on the vacuum gauge.

Degree of maximum attained vacuum in vacuum pump standard (reference) [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

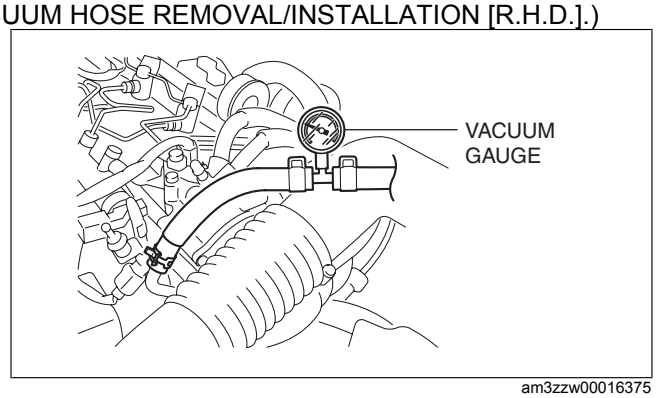
Measurement condition		Specification
Engine rotation speed	Measurement time	
While idling (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)	No time designation	93.3 kPa {700 mmHg, 27.6 inHg} or more

   (2) Verify that the measured vacuum matches the standard.
   - If they do not match, repair or replace the following malfunctioning locations.
     — Vacuum pump malfunction
     — Engine oil pressure decrease

SKYACTIV-D 1.5

1. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5])

2. Pinch open the clamp using pliers and disconnect the vacuum hose from the vacuum pump. (See VACUUM HOSE REMOVAL/INSTALLATION [L.H.D.].) (See VACUUM HOSE REMOVAL/INSTALLATION [R.H.D.].)
3. Install the vacuum gauge.
4. Warm up the engine.
5. Switch the ignition off.
6. Depress the brake pedal several times and decrease the vacuum of the vacuum gauge to **0 kPa {0 mmHg, 0 inHg}**.
7. Switch the ignition ON (engine on).



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Caution
- **If the brake pedal is released after switching the ignition ON (engine on), the vacuum in the vacuum pump may decrease and result in an incorrect reading when performing the inspection for the degree of vacuum in the vacuum pump. Depress and hold the brake pedal after switching the ignition ON (engine on).**

8. Perform the inspection for the degree of vacuum in the vacuum pump.
   (1) Measure the time from the measurement-start vacuum to the measurement-stop vacuum while taking readings on the vacuum gauge.

Degree of vacuum in vacuum pump maximum (reference) [SKYACTIV-D 1.5]

Measurement condition				Maximum value
Engine rotation speed	Brake pedal	Measurement-start vacuum	Measurement-stop vacuum	
While idling (See ENGINE TUNE-UP [SKYACTIV-D 1.5].)	Depressed	0 kPa {0 mmHg, 0 inHg}	67 kPa {503 mmHg, 20 inHg}	15 s + Engine completely warmed up

   (2) Verify if the measured time is the maximum value or less.
      - If it exceeds the maximum value, repair or replace the following malfunctioning locations.
        — Vacuum pump malfunction
        — Engine oil pressure decrease